Mechatronic Sensors Trainer

Fundamental Labs – Force Sensing Resistor

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QUANSER
INNOVATE · EDUCATE

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Force Resistor Lab

Why Explore Force Sensing Resistors?

Force sensing resistors measure the amount of force applied to the resistor. They can be used as pressure-sensitive user controls in touchpads and musical instruments. Force resistor arrays are used for measuring and analyzing pressure in medical applications such as prosthetics, podiatry, and patient monitoring. Understanding how to work within their limited sensing range and precision is essential to implementing force sensors effectively across a variety of applications.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**012_force_resistor**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Force Resistors**. The lab will guide you through the working principle of potentiometers, how to read and use them as well as using the datasheet to understand the sensor better.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have the provided force sensing resistor, with the appropriate connectors for the Sensors Trainer

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.

- You are familiar with the basics of the programming language you are using for this lab.

- Have your development environment ready,

    - If using Python, we recommend using Visual Studio Code.